

Manav Bhullar -- AI Portfolio

Interactive conversational portfolio. Visitors talk to an AI persona of Manav; answers are grounded in a Pinecone RAG knowledge base and rendered as React cards instead of a traditional multi-page site.

Next.js 15.2.8

React 19

Gemini Flash

Pinecone

Upstash Redis

PostHog

Codebase overview | 2026-08-20 | Source: Portfolio-main

Product idea

There is no persistent nav, sidebar, or About page. Me / Projects / Skills / Fun / Contact are conversation starters. The home hero collects a question and routes to /chat?query=...

1. What this is

The live product lives in Portfolio-main. A visitor lands on a single-column hero, types a question or taps a chip, and is sent into chat. From there the site is a conversation. The persona speaks in first person as Manavdeep Singh Bhullar -- Computer Engineering at Thapar Institute (TIET), Patiala, 2023-2027 -- working across full-stack, data analytics, and AI/ML.

The persona is explicitly not a general-purpose assistant: no writing code, no trivia, deflect back to the portfolio. Design is documented in DESIGN.md: warm tan surface, cream cards, organic cookie blobs behind clean rectangles, Google Sans Flex headlines, Inter body, forced light theme.

2. Visitor flow

1. Land /	Warm organic-blob hero. Search bar, five chips, welcome modal, GitHub star, Analytics button.
2. Ask /chat	Query auto-submits from the URL. Empty state offers a projects CTA; chips stay in the bottom bar.
3. Retrieve POST /api/chat	Rate limit, rewrite follow-ups, embed, Pinecone top-20, hybrid rerank, inject context.
4. Answer Chat UI	Gemini streams text plus at most one tool. Tools mount React cards. Citations and follow-up chips parse out of the text.

3. Pages and APIs

/	Landing hero. Forced light theme. Google Sans Flex display + Inter body.
/chat	Conversational UI via useChat (Vercel AI SDK). Shows the latest turn, not a full history thread.
/analytics	Private dashboard. Gated by ?key=manav or localStorage. Metrics from localStorage + PostHog.
POST /api/chat	Streaming LLM: gemini-flash-latest, maxDuration 60s, 10 req / 60s per IP.
/test, /test/title	Leftover dev pages, not part of the product.

4. The AI stack

Models

- Chat: gemini-flash-latest
- Follow-up rewrite: gemini-1.5-flash (last 6 messages rewritten into a standalone query)
- Embeddings: gemini-embedding-2, 3072 dimensions
- API keys picked at random from every GEMINI_API_KEY* / GOOGLE_API_KEY* env var

RAG pipeline

- Source of truth: src/lib/rag/knowledge-base.ts -- 20 self-contained documents
- Ingest: scripts/ingest.ts runs on every npm run build; SHA-256 cache in .rag-cache.json
- Store: Pinecone cosine, serverless AWS, 3072-d vectors
- Retrieve: embed query, Pinecone top 20, hybrid rerank 75% vector + 25% keyword/title
- Keep top 6 plus anything scoring 0.3 or higher
- Retrieved chunks merge into the single Gemini system message
- Persona prompt holds no bio data -- all facts come from RAG
- Retrieval failure is non-fatal: chat still runs without context

Persona rules

- First person as Manav; short punchy sentences; match the user's language
- Portfolio-only -- never write code, never answer general knowledge
- Must use retrieved numbers; must cite [citation: source_id]
- Every text reply ends with FOLLOW_UP_QUESTIONS (2-3 items)
- At most one tool per turn (maxSteps: 1)

5. Chat tools map to UI cards

Tool execute functions return a short string. The real payload is the client component ToolRenderer mounts from the tool name.

getPresentation	presentation.tsx -- name, degree, Patiala, three-domain bio
getProjects	AllProjects + carousel -- five cards; Floq/SCALES open interactive sandboxes
getSkills	skills.tsx -- four color-blocked groups: web, AI, data, soft skills
getResume	resume.tsx -- three PDF downloads: SDE, AI/ML, Data Analyst
getContact	contact.tsx -- email, phone, GitHub, LinkedIn, LeetCode, Tableau
getInterests	interests.tsx -- reading and fitness
getCrazy	crazy.tsx -- Gemini API key-rotation story (~7,500 req/day)

6. Featured projects

Floq -- green -- 4.8x speedup

Real-time ride-matching / carpooling. Node.js, PostgreSQL, Redis, Socket.io. FSM ride lifecycle, Redis SET NX PX plus Postgres FOR UPDATE SKIP LOCKED, constrained backtracking (40,320 to 2,520 sequences, 30% max detour), cache-aside driver GPS, 108 integration tests. Expanding the card opens a backtracking sandbox.

SCALES v3.0 -- orange -- CBTE pipeline

Automated short-answer grading. FastAPI, LiteLLM, HuggingFace. CBTE trust pipeline after v2 learned to trust its own hallucinations. Stages: CERA -> CGR -> CBTE -> Aggregator, tau = 0.5. Expanding the card opens a pipeline sandbox.

AI Portfolio RAG -- pink -- vector search

This site itself. Next.js, Pinecone, Gemini, Tailwind, Vercel AI SDK.

Olist Analytics -- blue -- 46K+ churned buyers

96,095 customers, RFM (48% Lost, 2% Loyal), late deliveries 2.5 vs 4.1 stars, DuckDB + Tableau Public dashboard.

NYC Taxi Analytics -- violet -- 9.38M records

9.38M raw trips to 540K cleaned, BigQuery surge windows, Folium choropleth of 263 NYC taxi zones.

The knowledge base also still describes PIP-RAG (Qdrant + Gemini over 226 companies, CGPA filters, key

rotation) and smaller MERN apps: Campus Marketplace, Housekeeping Management, Medicine Price Comparator. Those are in RAG, not on the carousel.

7. Person and knowledge base

Manavdeep Singh Bhullar -- B.E. Computer Engineering at TIET, Patiala, 2023-2027. Builds across full-stack, data analytics, and AI/ML. 400+ DSA problems. Technical Head of Virsa Society (ssavirsa.in rebuild, 20 sponsors, 4,000+ attendees). Deloitte data-analytics job simulation via Forage. Three role-specific resumes in /public: SDE, AI/ML, Data Analyst.

Indexed RAG topics (20 documents)

Background	about-me, achievements-certifications
Experience	Virsa Society, Deloitte Forage simulation
Projects	Floq (6 chunks), SCALES, PIP-RAG, Olist, NYC Taxi, other web
Skills	languages, web, AI, data, tools, soft skills
Personal	interests, why-hire-me, contact, crazy-hack

8. Quota protection and analytics

Quota

- Upstash sliding window 10 requests / 60s per IP; in-memory fallback for local
- Fails open if Redis errors so chat stays up
- 429 responses include Retry-After and X-RateLimit-* headers

Analytics

- PostHog: pageleave + chat_query_submitted with topic classification
- Vercel Analytics on the root layout
- localStorage ledger of queries (cap 500) and visits (cap 1000)
- Dashboard: visitors, query count, weekly traffic, topic bars, recent ledger
- Unlock via ?key=manav (also 1 / true). Cmd+Shift+A listener exists but is never mounted

9. Design system

#EDE6D6 surface	Page background (warm tan)
#FAF7EF card	Cards and chat surfaces
#191919 ink	Primary text / chips
#3FB37F Floq green	Focus ring, citations, Floq accent
#F0954A SCALES orange	Fun chip, SCALES accent
#E0559C PIP-RAG pink	Portfolio RAG accent
#3E8EDE Olist blue	Projects chip
#8B5FE0 NYC violet	Contact chip

Signature pattern: organic accent blob plus clean rectangular content. Pills, chips, and the search bar stay symmetric. Never a photorealistic face -- the MB mark is the identity treatment.

10. Tests and leftover agent work

tests/ is a custom four-tier harness (run-all.ts) covering rate limiting, RAG ingest/hash cache, chat route behavior, and some a11y/DOM checks, plus separate adversarial/empirical scripts. Vitest is in package.json but this runner is homemade. .agents/ is orchestration metadata from a prior mobile 100dvh + ESLint pass, not product.

11. Gaps worth knowing

Content drift

- Carousel replaced PIP-RAG with AI Portfolio RAG; knowledge base still indexes PIP-RAG
- Knowledge base still claims this site is in-memory RAG; it is Pinecone now
- Emails disagree: contact.tsx uses manavbhullar341@gmail.com; RAG uses manavbhullar2004@gmail.com

Leftovers from the original fork

- French Chargement du chat... on the /chat suspense fallback
- Unused getWeather tool and unused photos.tsx
- public/ still holds another author's screenshots (pharmassist, youbot, minishell, Toukoum logo)
- Welcome modal still says What's ???? / Why ???

Wiring

- AdminShortcutListener (Cmd+Shift+A) is never mounted in the layout
- Chip questions do not match client-side mock-tool shortcuts, so those quota shortcuts almost never fire
- ESLint is ignored during production builds (next.config.ts)
- OneDrive conflict copies exist (Dashboard / chat Manav's MacBook Air)

Bottom line

The site is a designed chat product that is the portfolio. The interesting engineering is the RAG + tool-card loop, not a gallery of static pages. The person it represents is a TIET Computer Engineering student who wants to be evaluated on Floq, SCALES, and the analytics work -- and this repo is both the brochure and one of the projects.

Highest-leverage cleanup

Align RAG vs carousel vs contact details, and strip the fork leftovers so a recruiter only sees Manav's work.